

Atrial Fibrillation Biomarker Prediction to the Personalization of Electrophysiological Models

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Abstract

Atrial fibrillation (AF) is the most common cardiac arrhythmia, but current treatments show limited efficacy due to interpatient variability. Personalized computational models could improve therapy selection, yet their use is constrained by computational cost and time. We propose a machine learning (ML) framework to rapidly personalize atrial electrophysiological models.

A virtual cohort of 2,100 ionic profiles was generated and evaluated through rotor-based simulations to assess reentry sustainability and extract biomarkers. ML models were trained to classify reentry-capable profiles and predict biomarker values directly from ionic current configurations.

The classifiers achieved F1-scores of 0.95, while regression models predicted biomarkers with R^2 up to 0.97. In a case study, the framework instantly identified patient-specific ionic profiles among 100,000 candidates, a task that would require ~625 days using conventional simulations.

ML-based workflow drastically reduces the time required to generate patient-specific cardiac models. By estimating complex electrophysiological behaviors within seconds, it enables the integration of digital twins into clinical research and supports the design of personalized therapies for AF.

1. Introduction

Atrial fibrillation (AF) is the most common cardiac arrhythmia, affecting a significant portion of the population, with incidence increasing with age. While its prevalence is estimated at 1–2% in the general population, it can reach up to 5% in patients over 65 years. AF represents a complex clinical challenge, as the mechanisms that initiate and sustain the arrhythmia are not fully understood [1].

Currently, there are two main approaches for AF treatment: pharmacological therapy with antiarrhythmic drugs and catheter ablation. However, both show

considerable recurrence rates, highlighting the need for more effective strategies. Reported recurrence within the first year is 59% with drug therapy and 36% with catheter ablation, clearly evidencing the limitations of current treatments.

This lack of efficacy is partly explained by the fact that the pathophysiological mechanisms of AF vary among patients. Consequently, a treatment that is effective for one individual may fail in another.

To overcome this limitation, the development of an individualized “digital twin” of each patient’s heart has emerged as a promising solution [2]. Such personalized computational models can simulate cardiac response to different treatments, enabling a more precise and patient-tailored therapy selection [3].

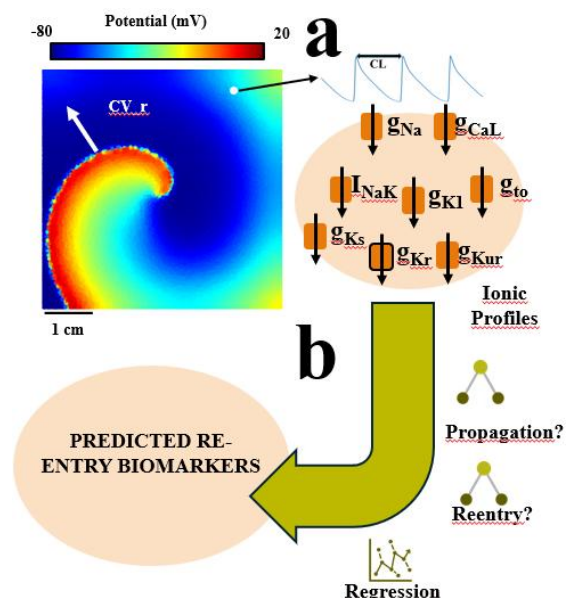


Figure 1. Methodology employed. (a) Different ionic profiles were simulated under reentry events, and reentry biomarkers (CV_r and CL) were measured. (b) Machine learning models were trained and evaluated to assess which ionic profiles propagated [6] and reentered, and finally to predict biomarkers.

The main barrier to implementing digital twins lies in the high computational cost and time required for model personalization. This process is so resource-intensive that it severely limits both clinical and research applications. Therefore, new methodologies are essential to drastically reduce the time required to generate these personalized models without compromising accuracy, ensuring their integration into clinical practice and improving patient outcomes.

In this work, we develop artificial intelligence (AI) tools to accelerate the estimation of AF biomarkers -cycle length (CL) and reentrant conduction velocity (CV_r)- with the aim of assigning each patient and atrial region the most appropriate electrophysiological model.

2. Methods

2.1. Model population generation

We started from the Koivumäki electrophysiological model [4], which mathematically represents the electrical activity in cardiac cells. This model, originally designed for sinus rhythm, was modified to represent chronic atrial fibrillation (cAF) by altering specific ionic conductances.

To model physiological variability among patients, a virtual population of 2,100 ionic profiles was created. Eight key ionic currents (see *Figure 1*) were selected to be varied within a range of -75% to +150%. To ensure a uniform representation of the space of possible combinations, the Latin hypercube sampling (LHS) method was employed as done in [5].

2.2. Simulations and biomarker extraction

Each of the 2,100 ionic profiles was evaluated through electrophysiological simulations to analyze their ability to propagate the action potential and, if so, their ability to sustain reentrant activity. In these simulations, a rotor was induced to determine this behavior, and key biomarkers were measured (as shown in *Figure 1A*), namely: CL and CV_r.

The CL is the time elapsed between two consecutive activations at a specific point. The CV_r measures the speed at which the electrical wavefront propagates along the reentrant pathway. Both biomarkers were measured directly in the rotor simulations to determine the ability of each ionic profile to sustain reentrant activity [6].

2.3. Machine learning model training and evaluation

A series of machine learning (ML) models were trained to classify the ionic profiles capable of maintaining reentry and to predict biomarker values (CV_r and CL) directly from ionic current configurations. The models were trained

using hyperparameter search (grid search) and cross-validation techniques. The models and their prediction errors were evaluated using standard metrics: F1-score for classification, and MAE, MSE, and R² for regression.

3. Results

3.1. Analysis of electrophysiological properties

Figure 2 shows significant differences in ionic conductances between profiles capable of sustaining reentry and those that could not, as assessed using the Mann-Whitney U test. The correlation of these currents with biomarkers in reentrant models is also presented.

Interestingly, I_{Na} displayed a strong correlation with reentry biomarkers but did not show significant differences between reentrant and non-reentrant models. In contrast, I_{K1} exhibited the strongest correlation with CL and showed significant differences between reentrant and non-reentrant models.

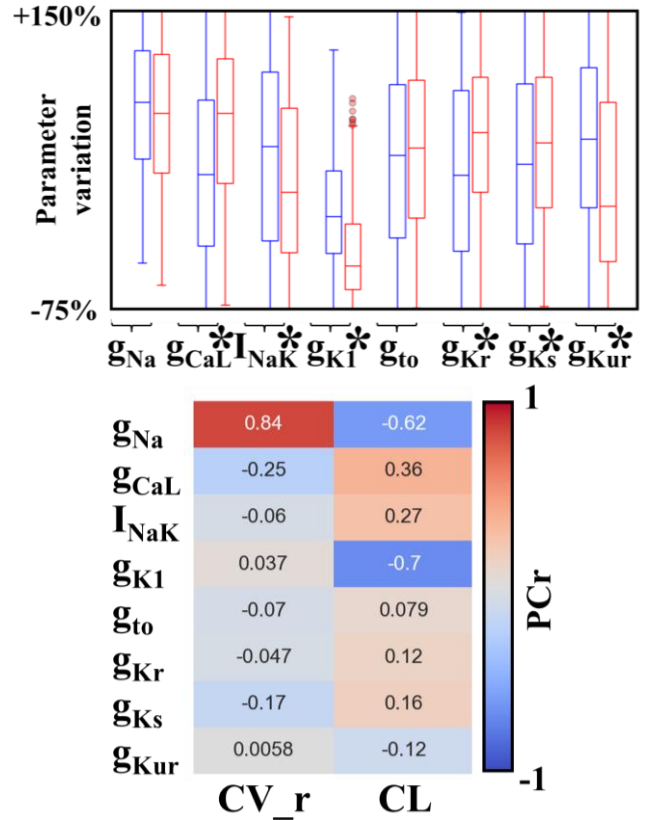


Figure 2. (Top) Boxplots comparing ionic current conductances between reentrant (blue) and non-reentrant (red) models. Significant differences are marked * (p < 0.05). (Bottom) Partial correlation (PCr) between ionic currents and reentry biomarkers. Cycle length is dependent on various biomarkers.

Figure 3 illustrates the correlation between reentry biomarkers and action potential propagation biomarkers measured in [6]. This analysis revealed a high correlation between action potential and reentry biomarkers.

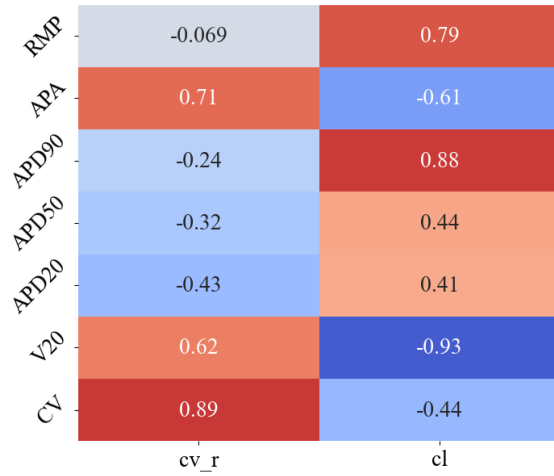


Figure 3. Relationship between reentry biomarkers and 1 Hz pacing biomarkers [6].

3.2. Evaluation of ML models

Figure 4 summarizes the performance of the machine learning models. Different models were evaluated, with the best results in all three stages—conduction [6] or reentry classification and biomarker regression—being obtained with multilayer perceptrons with between 3 and 4 dense layers. A confusion matrix highlights the classifier’s results. In classification tasks, an F1-score of 0.95 was achieved for reentry detection, indicating high reliability. In regression tasks, the models predicted reentry biomarkers with an R^2 value of up to 0.969.

3.3. Case study

To explore the clinical applicability of the developed workflow, we considered a patient case with experimental measurements of $CV_r = 337$ mm/s and $CL = 185$ ms. This reflects common clinical practice, where some biomarkers can be measured, ionic currents are inaccessible.

Based on these measured values, the trained ML model was used to infer which ionic configurations, within a set of 100,000 virtual models, could propagate, sustain reentry, and present biomarker values closest to the patient’s data. These configurations represented a wide range of physiological and pathological conditions and were evaluated without the need for new simulations in this specific case.

The system identified a subset of ionic models with an electrophysiological profile consistent with the patient’s biomarkers. These models could then be used for further simulations to assess potential therapies in silico.

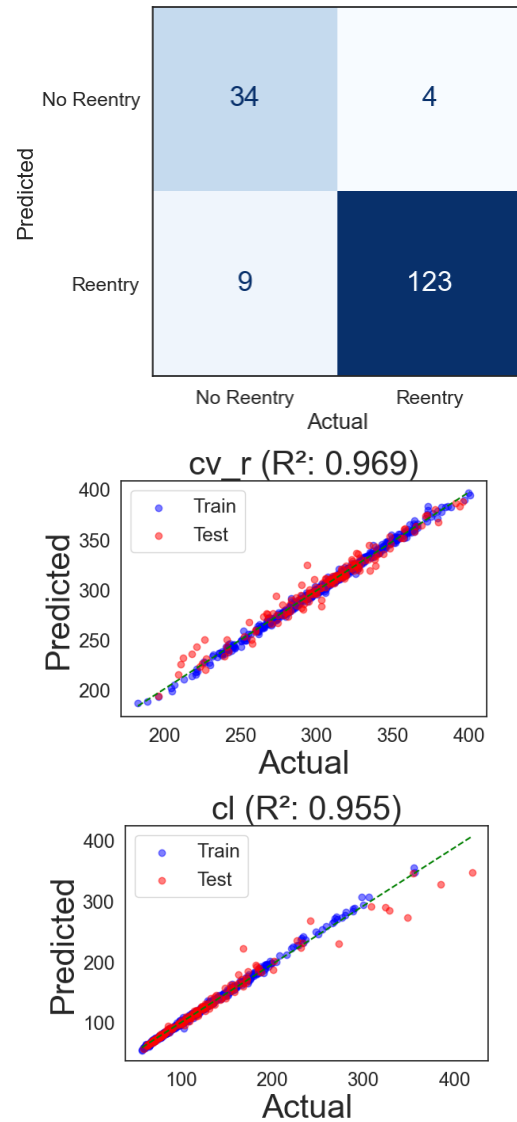


Figure 4. Predictions. (Top) Confusion matrix identifying reentrant models, prioritizing sensitivity over specificity. (Bottom) Predicted reentry values for training and test datasets.

Figure 5 shows the ionic current distributions for the 20 ionic models with predicted biomarkers closest to the patient’s values. Among these closest models, the mean CV_r was 336.86 ± 1.86 mm/s, and the mean CL was 185.17 ± 1.67 ms. This result indicates that different ionic models can yield similar biomarker values, though they may respond differently to pharmacological treatments. Notably, the I_{Na} and I_{K1} currents—shown throughout this study to have the largest impact on biomarkers—exhibited the lowest variability at this stage.

Finally, simulating all 100,000 reentries on the simulation platform would have required 625 days (9 minutes per simulation), whereas the ML-based estimation was practically instantaneous.

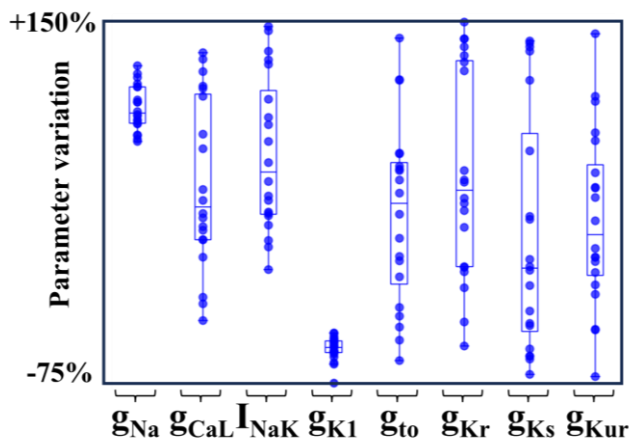


Figure 5. Distribution of ionic currents for the 20 ionic models, out of 100,000, whose reentry biomarkers best matched the biomarkers measured in a patient [7].

4. Discussion

This work successfully achieved its main objective: to develop and validate a machine learning-based methodology capable of drastically accelerating the creation of cardiac digital twins. The proposed platform overcomes the primary barrier to the clinical adoption of personalized models: the high computational and time costs of electrophysiological simulations.

The key innovation lies in the system's ability to instantaneously estimate complex biomarkers, such as conduction velocity and reentrant circuit characteristics, across tens of thousands of ionic profiles. This represents a paradigm shift in efficiency: a characterization process that would traditionally require 625 days of simulations can now be completed in seconds. By removing this computational bottleneck, the framework enables the exploration of physiological and pathological diversity at an unprecedented scale.

By decoupling prediction from simulation, the method rapidly identifies ionic profiles consistent with a patient's clinical biomarkers. This not only accelerates the construction of personalized models but also provides deeper insights, showing that different ionic configurations can yield similar electrical behaviors. Such findings reinforce the importance of patient-specific models when evaluating, for instance, pharmacological treatments.

In conclusion, this work establishes a robust and efficient methodological foundation that accelerates the generation of cardiac digital twins. By transforming a process of months into an instantaneous task, it takes a decisive step toward making personalized simulation a viable tool in clinical research, and in the future, a cornerstone for risk stratification and tailored therapy design in cardiology.

Acknowledgments

This work was funded by the Generalitat Valenciana grant AICO/2021/318 (Consolidables 2021), and by grants PID2020-114291RB-I00, PID2023-148702OB-I00 and EraNet PCI2024-153442 funded by MCIN/10.13039/501100011033 and by "ERDF A way of making Europe."

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